Name: __________________________ Class: ____________ Date: ____________ **ID: A**

Chemistry Chapter 17 Practice Quiz 1 — Equilibrium Fundamentals

Multiple Choice

Identify the choice that best completes the statement or answers the question.

___ 1. Chemical equilibrium is reached when

a. all reactants have been converted to products

b. the forward and reverse reaction rates are equal and concentrations are stable

c. the concentrations of reactants and products are equal

d. the reaction has stopped completely

___ 2. Which of the following is an example of a reversible reaction?

a. Burning wood in a fireplace

b. The Haber process: $N_2 + 3H_2 \rightleftharpoons 2NH_3$

c. Exploding dynamite

d. Dissolving NaCl in water (one-way only)

___ 3. For the reaction $aA + bB \rightleftharpoons cC + dD$, the Keq expression is

a. $Keq = [A]^a[B]^b / [C]^c[D]^d$

b. $Keq = [C]^c[D]^d / [A]^a[B]^b$

c. $Keq = (c+d) / (a+b)$

d. $Keq = [C][D] / [A][B]$

___ 4. Which of the following is NOT included in a Keq expression?

a. Gases

b. Aqueous solutions

c. Pure solids and pure liquids

d. Dissolved ions

___ 5. A reaction has Keq = 650. Which statement is correct?

a. The reaction barely proceeds; mostly reactants remain

b. The reaction proceeds almost to completion; mostly products are present

c. Reactants and products are present in a 1:1 ratio

d. The reverse reaction predominates

True / False

Indicate whether the statement is true or false.

___ 6. In a reversible reaction, both reactants and products exist together as a mixture.

___ 7. At equilibrium, the concentrations of reactants and products must be equal.

___ 8. The equilibrium position depends on the conditions at which equilibrium is reached.

___ 9. A $Keq < 1$ means the forward reaction predominates and mostly products are present.

___ 10. The numerical value of Keq must be determined experimentally.

___ 11. An irreversible reaction can occur in both the forward and reverse directions.

___ 12. Homogeneous equilibrium involves substances that are all in the same state of matter.

___ 13. Pure liquids are included in the Keq expression because they have a measurable concentration.

___ 14. For a reaction at equilibrium, the forward and reverse reaction rates are equal.

___ 15. A Keq > 1 means the equilibrium lies to the right and [products] > [reactants].

Short Answer

16. Write the Keq expression for the following reaction: $3H_2(g) + N_2(g) \rightleftharpoons 2NH_3(g)$

17. Explain the difference between a reversible reaction and an irreversible reaction. Give one real-world example of each.

18. A student finds Keq = 0.0025 for a reaction. What does this tell you about the equilibrium position? Is the reaction product-favored or reactant-favored? Explain.

Essay

Answer TWO of the following. You may answer all 3 for possible bonus.

19. Explain what chemical equilibrium means. In your explanation, address: (a) what happens to the forward and reverse reaction rates at equilibrium, (b) whether concentrations stop changing, (c) whether the forward and reverse reactions stop occurring.

20. Describe the law of chemical equilibrium and explain how the Keq expression is written from a balanced chemical equation. Include the rule for coefficients and the rule for pure solids/liquids. Give a worked example using a reaction of your choice.

21. Compare and contrast homogeneous equilibrium and heterogeneous equilibrium. Give one example of each and write the Keq expression for each. Explain why solids and liquids do not appear in the Keq expression.